

Regresní analýza

LMER model - fixní ticker

Všechny fixní efekty mimo modelu jsou statisticky významné. Velmi malý efekt náhodné složky (měsíc sběru). Nízké mezní R^2 , nelze vypočítat podmíněné R^2 kvůli nízkému efektu měsíců.

```
subset_args <- c(
  "S",
  #"moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model",
  "ticker"
)
interactions <- c(
  #"(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)

## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
summary(rmr_relative)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula
## Data: train_data_rel
##
## REML criterion at convergence: -1766596
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.9817 -0.3327  0.0635  0.4441  5.1644
##
## Random effects:
## Groups           Name          Variance Std.Dev.
## snapshot_month (Intercept) 5.208e-07 0.0007216
## Residual                2.746e-04 0.0165700
## Number of obs: 329512, groups: snapshot_month, 4
```

```
##
## Fixed effects:
##      Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -3.113e-02  1.331e-03  5.360e+02 -23.389 < 2e-16 ***
## S            1.243e-05  6.012e-07  8.341e+04  20.684 < 2e-16 ***
## K           -7.506e-06  1.395e-07  3.295e+05 -53.819 < 2e-16 ***
## r            1.781e+00  2.844e-02  1.414e+05  62.616 < 2e-16 ***
## sigma       -8.096e-02  1.256e-03  5.932e+04 -64.461 < 2e-16 ***
## q           -4.741e-01  4.962e-02  3.294e+05  -9.555 < 2e-16 ***
## T            3.880e-03  4.146e-05  3.294e+05  93.593 < 2e-16 ***
## am_eueuropean -3.048e-02  1.497e-03  4.768e+04 -20.366 < 2e-16 ***
## call_puts    -3.148e-03  6.029e-05  3.295e+05 -52.215 < 2e-16 ***
## modeluser_crr -8.143e-06  7.071e-05  3.295e+05  -0.115  0.9083
## modeluser_mc  1.578e-05  7.070e-05  3.295e+05   0.223  0.8234
## ticker~XDA    -1.496e-02  2.345e-03  2.352e+05  -6.377 1.80e-10 ***
## ticker~XDB     1.869e-02  2.161e-03  2.249e+05   8.649 < 2e-16 ***
## ticker~XDE     4.192e-03  1.882e-03  1.795e+05   2.228  0.0259 *
## ticker~XDN     1.217e-02  2.013e-03  1.871e+05   6.046 1.49e-09 ***
## tickerAAPL    -1.375e-02  4.226e-04  1.776e+05 -32.535 < 2e-16 ***
## tickerAMD      1.059e-02  1.749e-04  1.085e+05  60.537 < 2e-16 ***
## tickerAMZN    -4.600e-03  3.216e-04  1.833e+05 -14.304 < 2e-16 ***
## tickerGOOGL    7.413e-04  4.072e-04  1.591e+05   1.820  0.0687 .
## tickerJNJ      5.889e-04  1.740e-03  3.238e+05   0.338  0.7350
## tickerKO      -3.924e-03  1.585e-03  3.246e+05  -2.475  0.0133 *
## tickerMETA    -7.866e-03  3.447e-04  6.645e+04 -22.817 < 2e-16 ***
## tickerMSFT    -6.975e-03  5.708e-04  1.612e+05 -12.219 < 2e-16 ***
## tickerNFLX    -1.856e-03  3.399e-04  2.539e+05  -5.460 4.77e-08 ***
## tickerNVDA     3.693e-03  2.449e-04  2.861e+05  15.077 < 2e-16 ***
## tickerPG      -2.851e-03  1.437e-03  3.222e+05  -1.985  0.0472 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

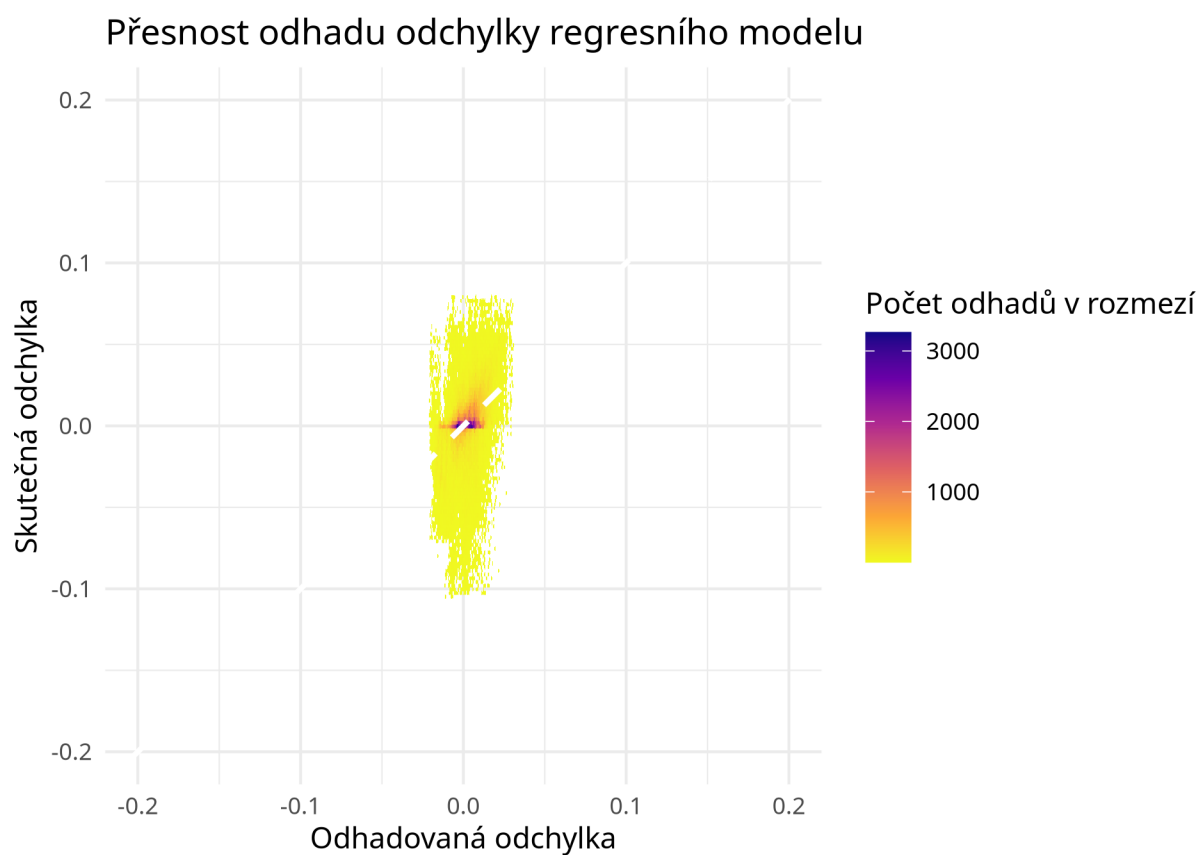
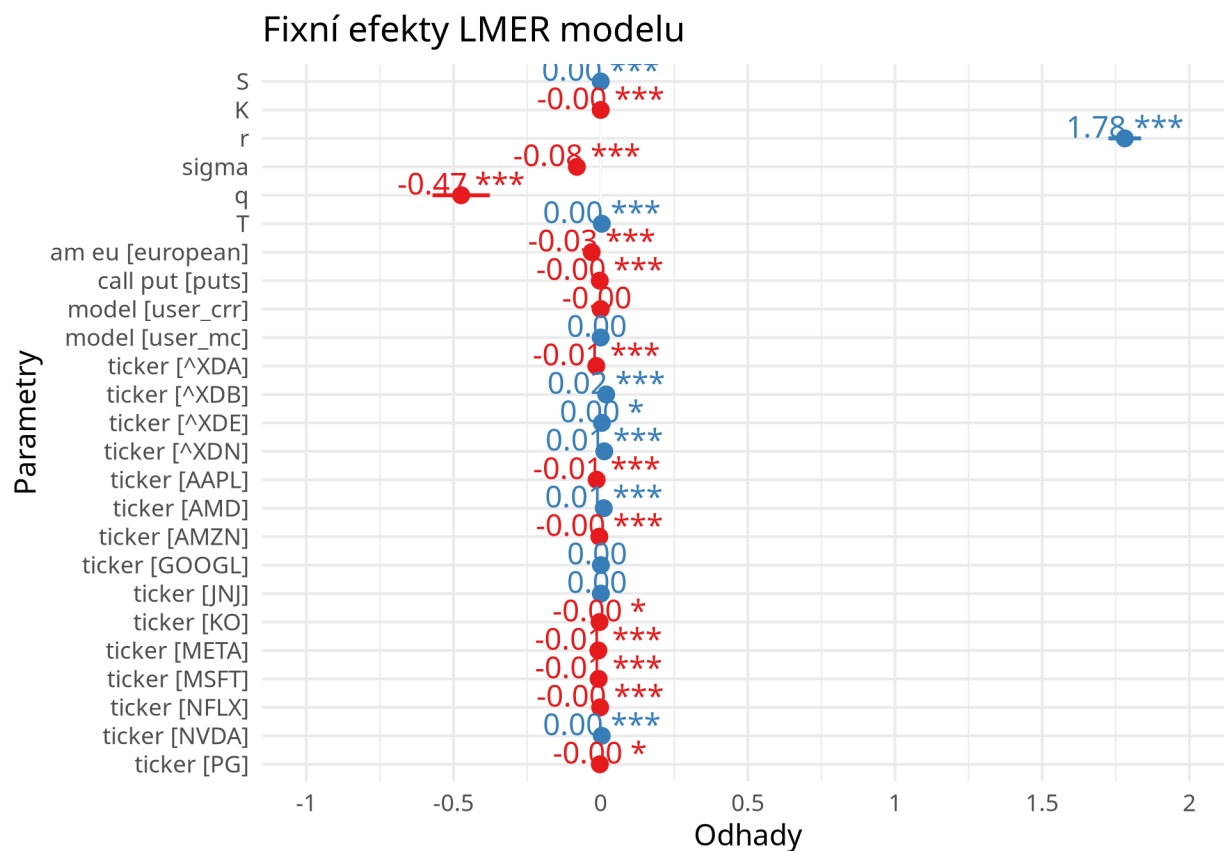
##
## Correlation matrix not shown by default, as p = 26 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.176
```



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF    F value Pr(>F)
## S           0.1175  0.11746     1   83414   427.8228 <2e-16 ***
## K           0.7953  0.79528     1  329480  2896.5112 <2e-16 ***
## r           1.0765  1.07650     1 141352  3920.7494 <2e-16 ***
## sigma       1.1409  1.14089     1   59322  4155.2666 <2e-16 ***
## q           0.0251  0.02507     1  329356    91.2919 <2e-16 ***
## T           2.4051  2.40508     1  329399  8759.6239 <2e-16 ***
## am_eu       0.1139  0.11388     1   47677   414.7580 <2e-16 ***
## call_put    0.7486  0.74858     1  329483  2726.4416 <2e-16 ***
## model       0.0000  0.00002     2  329483    0.0592  0.9425
## ticker     4.6973  0.31315    15 262975  1140.5449 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $snapshot_month
##      (Intercept)
## 01 -7.129052e-04
## 02  9.922789e-04
## 03 -5.833745e-05
## 05 -2.210362e-04
##
## with conditional variances for "snapshot_month"
```

LMER model - náhodný ticker

Mnohem lepší R^2 , stále malé. Nevýznamný poměr S/K a styl využití opce (a model).

```
subset_args <- c(
  "#S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "model"
  #"ticker"
)
interactions <- c(
  "(1 | ticker)",
  "(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(rmr_relative)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula
## Data: train_data_rel
##
## REML criterion at convergence: -1766277
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0098 -0.3318  0.0604  0.4445  5.1863
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## ticker          (Intercept) 9.816e-05 0.0099075
## snapshot_month (Intercept) 6.710e-07 0.0008192
## Residual                2.749e-04 0.0165805
## Number of obs: 329512, groups: ticker, 17; snapshot_month, 4
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -3.043e-02  3.140e-03 2.101e+01 -9.691 3.33e-09 ***
## moneyness    9.876e-05  3.307e-05 3.285e+05  2.987  0.00282 **
## K            -6.864e-06  1.596e-07 3.179e+05 -43.008 < 2e-16 ***
## r            1.765e+00  2.846e-02 1.847e+05  62.027 < 2e-16 ***
## sigma       -7.654e-02  1.238e-03 1.113e+05 -61.801 < 2e-16 ***
## q           -5.477e-01  4.840e-02 7.261e+03 -11.316 < 2e-16 ***
## T            3.863e-03  4.169e-05 3.295e+05  92.647 < 2e-16 ***
## am_eueuropean -1.984e-02  5.334e-03 1.519e+01 -3.719  0.00202 **
## call_putpts  -3.112e-03  6.034e-05 3.295e+05 -51.571 < 2e-16 ***
## modelusr_crr -9.530e-06  7.075e-05 3.295e+05 -0.135  0.89285
## modelusr_mc   1.392e-05  7.075e-05 3.295e+05  0.197  0.84407
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) mnyss K      r      sigma q      T      am_rpn cll_pt
## moneyness    -0.020
## K            -0.016  0.500
## r            -0.334  0.019  0.007
## sigma        -0.138 -0.057 -0.069  0.002
## q            -0.143  0.011  0.005 -0.026  0.048
## T            -0.058 -0.100 -0.075  0.157 -0.007  0.001
## am_eueuropn  -0.508 -0.005 -0.012 -0.002  0.057  0.088  0.003
## call_putpts  -0.006 -0.038  0.211 -0.011 -0.014  0.000 -0.018 -0.002
## modelsr_crr  -0.011  0.000  0.000  0.000  0.000 -0.001  0.000  0.000  0.000
## modelusr_mc  -0.011 -0.001 -0.001  0.000  0.001 -0.001  0.000  0.000  0.001
##
##              mdlr_c
## moneyness
```

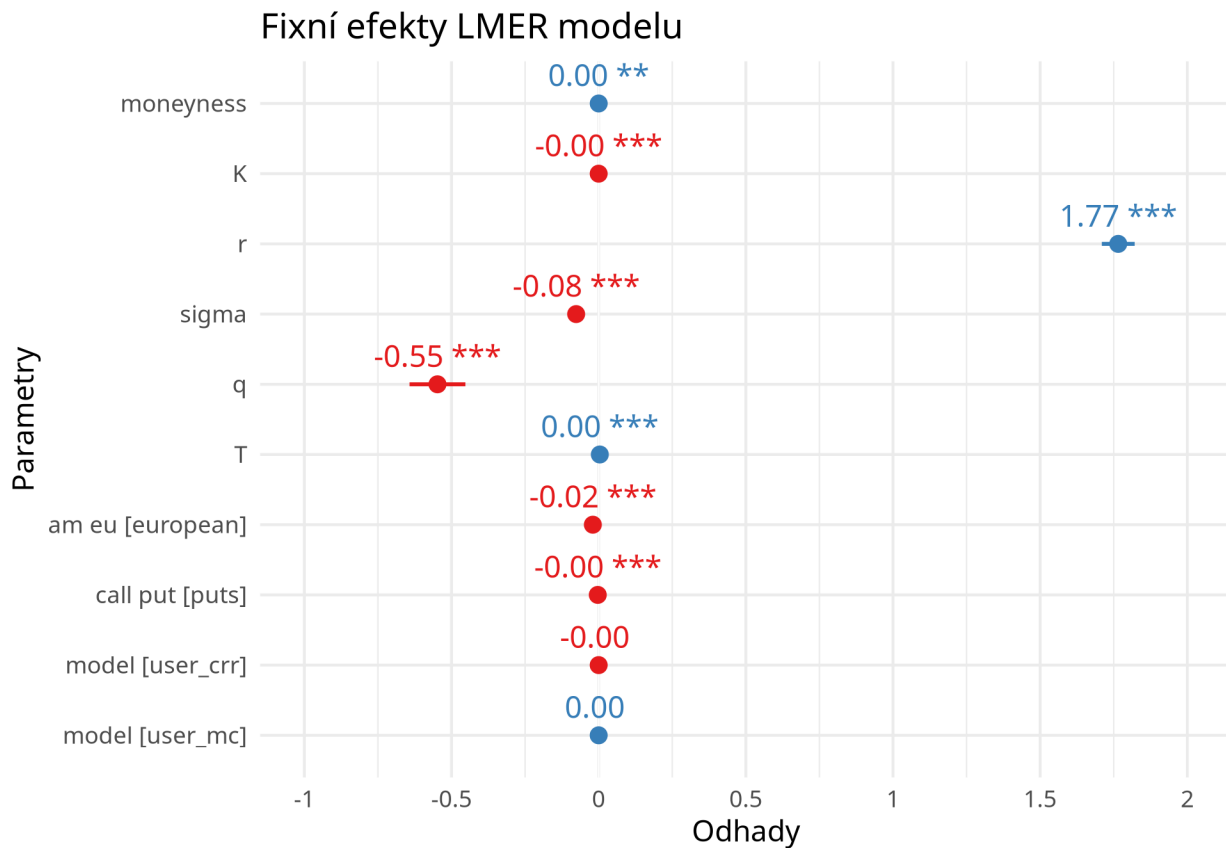
```
## K
## r
## sigma
## q
## T
## am_eueuropn
## call_putpts
## modelsr_crr
## modelusr_mc 0.500
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
r2(rmr_relative)
```

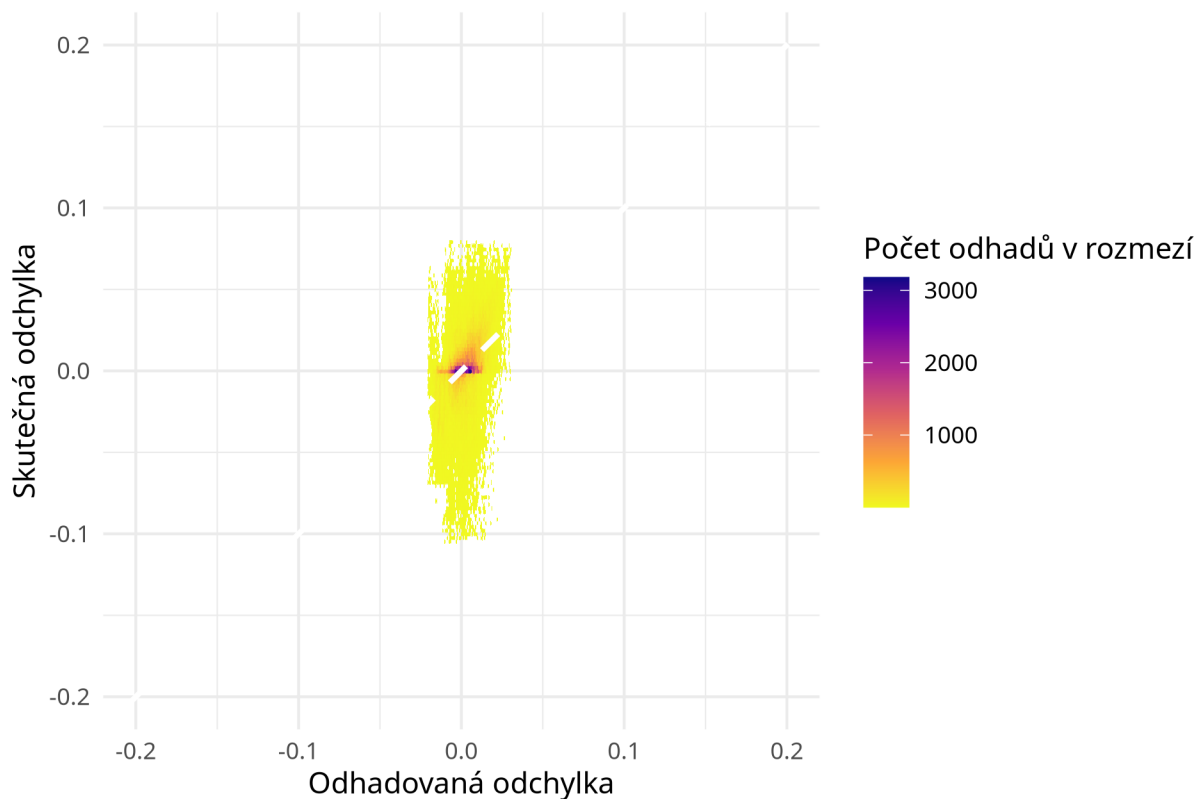
```
## Warning: Can't compute random effect variances. Some variance components equal
## zero. Your model may suffer from singularity (see `?lme4::isSingular`
## and `?performance::check_singularity`).
## Decrease the `tolerance` level to force the calculation of random effect
## variances, or impose priors on your random effects parameters (using
## packages like `brms` or `glmmTMB`).

## Random effect variances not available. Returned R2 does not account for random effects.

## # R2 for Mixed Models
##
## Conditional R2: NA
## Marginal R2: 0.321
```



Přesnost odhadu odchylny regresního modelu



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF    F value    Pr(>F)
## moneyness 0.00245 0.00245     1 328514     8.9215 0.002818 **
## K          0.50850 0.50850     1 317863 1849.6632 < 2.2e-16 ***
## r          1.05768 1.05768     1 184707 3847.3058 < 2.2e-16 ***
## sigma      1.04999 1.04999     1 111346 3819.3525 < 2.2e-16 ***
## q          0.03520 0.03520     1   7261  128.0488 < 2.2e-16 ***
## T          2.35972 2.35972     1 329452 8583.5268 < 2.2e-16 ***
## am_eu      0.00380 0.00380     1    15    13.8308 0.002017 **
## call_put   0.73116 0.73116     1 329485 2659.6153 < 2.2e-16 ***
## model      0.00003 0.00002     2 329483     0.0555 0.945970
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $ticker
##      (Intercept)
## ^RUT    0.0198248629
## ^XDA   -0.0246242852
## ^XDB    0.0088074687
## ^XDE   -0.0055644258
## ^XDN    0.0015563796
## AAPL   -0.0117662628
## AMD     0.0110811235
## AMZN   -0.0036266622
```

```
## GOOGL 0.0033001023
## JNJ 0.0047768113
## KO -0.0017662644
## META -0.0023118431
## MSFT -0.0028877790
## NFLX -0.0025451289
## NVDA 0.0038170048
## PG -0.0001604142
## TSLA 0.0020893125
##
## $snapshot_month
## (Intercept)
## 01 -0.0006664638
## 02 0.0008346210
## 03 -0.0007334943
## 05 0.0005653371
##
## with conditional variances for "ticker" "snapshot_month"
```

LMER model - náhodný ticker (fixní efekt měsíce sběru)

Stejně významnosti jako model s náhodným efektem měsíce, menší mezní R^2 než u předchozího modelu, podmíněné R^2 je větší než mezní R^2 předchozího modelu.

```
subset_args <- c(
  #"S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
  "call_put",
  "snapshot_month",
  "model"
  #"ticker"
)
interactions <- c(
  "(1 | ticker)"
  #"(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(rmr_relative)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```

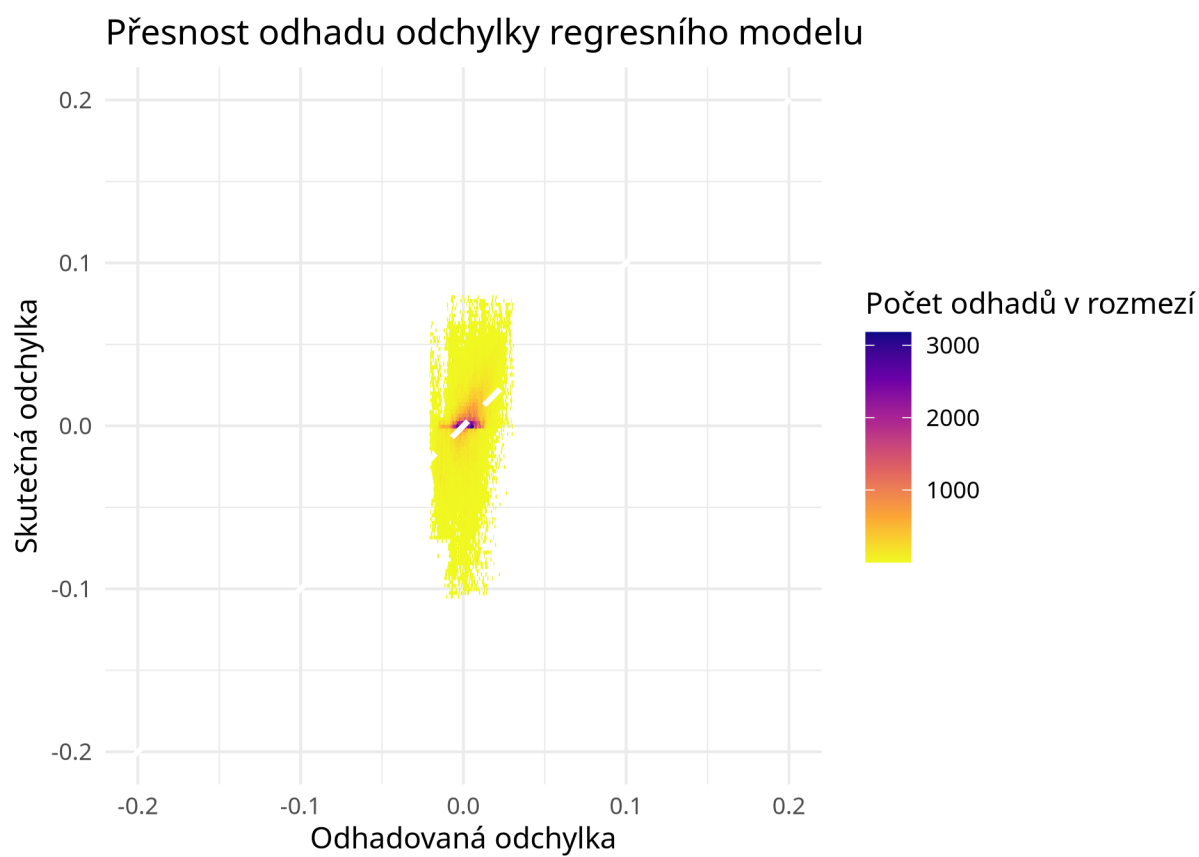
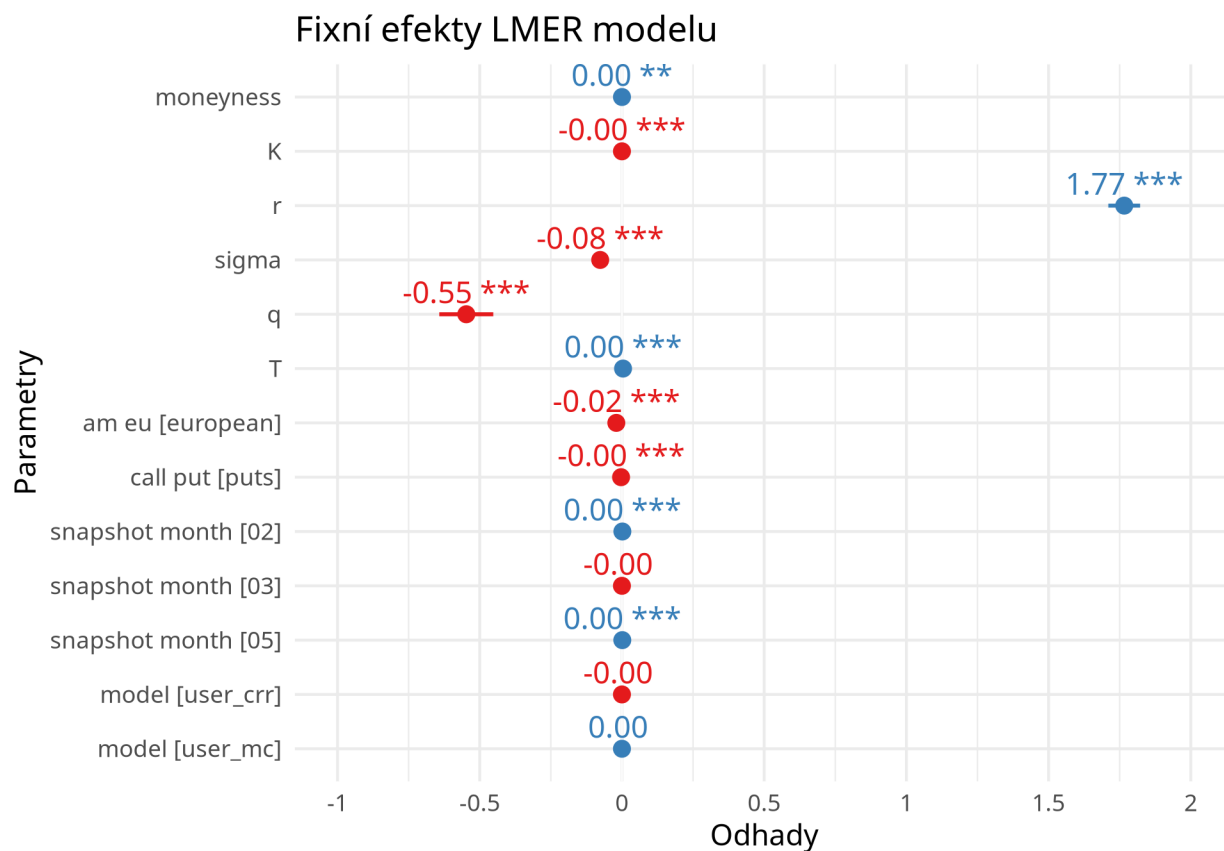
## lmerModLmerTest]
## Formula: formula
##   Data: train_data_rel
##
## REML criterion at convergence: -1766244
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0101 -0.3318  0.0604  0.4445  5.1860
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   ticker   (Intercept) 9.812e-05 0.009906
##   Residual                2.749e-04 0.016581
## Number of obs: 329512, groups:  ticker, 17
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -3.114e-02  3.112e-03 2.036e+01 -10.006 2.62e-09 ***
## moneyness    9.848e-05  3.307e-05 3.290e+05  2.978  0.00290 **
## K            -6.866e-06  1.596e-07 3.190e+05 -43.015 < 2e-16 ***
## r            1.766e+00  2.850e-02 3.295e+05  61.965 < 2e-16 ***
## sigma       -7.649e-02  1.241e-03 2.776e+05 -61.636 < 2e-16 ***
## q           -5.473e-01  4.840e-02 7.259e+03 -11.306 < 2e-16 ***
## T            3.863e-03  4.169e-05 3.295e+05  92.653 < 2e-16 ***
## am_european -1.982e-02  5.333e-03 1.520e+01  -3.717  0.00203 **
## call_puts    -3.112e-03  6.034e-05 3.295e+05 -51.574 < 2e-16 ***
## snapshot_month02 1.509e-03  8.419e-05 3.291e+05  17.925 < 2e-16 ***
## snapshot_month03 -6.770e-05  8.925e-05 3.295e+05  -0.759  0.44814
## snapshot_month05 1.240e-03  1.090e-04 3.262e+05  11.373 < 2e-16 ***
## modeluser_crr -9.542e-06  7.075e-05 3.295e+05  -0.135  0.89272
## modeluser_mc  1.391e-05  7.075e-05 3.295e+05   0.197  0.84410
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 14 > 12.
## Use print(x, correlation=TRUE) or
##   vcov(x)           if you need it

## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## # R2 for Mixed Models
##
##   Conditional R2: 0.454
##   Marginal R2: 0.259

```



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF    F value    Pr(>F)
## moneyness    0.00244  0.00244     1 328987    8.8704  0.002899 **
## K            0.50867  0.50867     1 319024 1850.2913 < 2.2e-16 ***
## r            1.05558  1.05558     1 329496 3839.6935 < 2.2e-16 ***
## sigma        1.04439  1.04439     1 277623 3798.9940 < 2.2e-16 ***
## q            0.03514  0.03514     1   7259  127.8279 < 2.2e-16 ***
## T            2.36000  2.36000     1 329490 8584.5360 < 2.2e-16 ***
## am_eu        0.00380  0.00380     1    15   13.8132  0.002026 **
## call_put     0.73122  0.73122     1 329483 2659.8382 < 2.2e-16 ***
## snapshot_month 0.14439  0.04813     3 327415  175.0688 < 2.2e-16 ***
## model        0.00003  0.00002     2 329483    0.0556  0.945932
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $ticker
##              (Intercept)
## ^RUT    0.0198232763
## ^XDA   -0.0246245864
## ^XDB    0.0088089399
## ^XDE   -0.0055634411
## ^XDN    0.0015558108
## AAPL   -0.0117619301
## AMD     0.0110707091
## AMZN   -0.0036223174
## GOOGL   0.0033037797
## JNJ     0.0047728753
## KO     -0.0017680271
## META   -0.0023104190
## MSFT   -0.0028828390
## NFLX   -0.0025405844
## NVDA    0.0038173705
## PG     -0.0001614436
## TSLA    0.0020828256
##
## with conditional variances for "ticker"
```

LMER model - náhodný ticker (žádný efekt měsíce)

Všechny parametry statisticky významné, lepší R^2 než u předchozího modelu. Mezní R^2 stále horší než u modelu s náhodným efektem měsíce.

```
subset_args <- c(
  #"S",
  "moneyness",
  "K",
  "r",
  "sigma",
  "q",
  "T",
  "am_eu",
```

```

"call_put",
"model"
#"ticker"
)
interactions <- c(
  "(1 | ticker)"
  #"(1 | snapshot_month)"
)

rhs <- paste(c(subset_args, interactions), collapse = " + ")
formula <- as.formula(paste("pricing_error ~ ", rhs))

rmr_relative <- lmer(formula, data = train_data_rel)

## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.

summary(rmr_relative)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: formula
## Data: train_data_rel
##
## REML criterion at convergence: -1765770
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -6.9596 -0.3304 0.0579 0.4389 5.2279
##
## Random effects:
## Groups Name Variance Std.Dev.
## ticker (Intercept) 0.0001032 0.01016
## Residual 0.0002753 0.01659
## Number of obs: 329512, groups: ticker, 17
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) -2.251e-02 3.146e-03 1.930e+01 -7.153 7.73e-07 ***
## moneyness 1.383e-04 3.284e-05 3.290e+05 4.210 2.55e-05 ***
## K -6.594e-06 1.579e-07 3.188e+05 -41.759 < 2e-16 ***
## r 1.636e+00 2.416e-02 3.286e+05 67.724 < 2e-16 ***
## sigma -8.373e-02 9.635e-04 3.072e+05 -86.906 < 2e-16 ***
## q -6.303e-01 4.830e-02 7.947e+03 -13.050 < 2e-16 ***
## T 3.807e-03 4.160e-05 3.295e+05 91.520 < 2e-16 ***
## am_eueuropean -2.227e-02 5.463e-03 1.518e+01 -4.077 0.000969 ***
## call_putputs -3.091e-03 6.037e-05 3.295e+05 -51.194 < 2e-16 ***
## modeluser_crr -7.615e-06 7.081e-05 3.295e+05 -0.108 0.914356
## modeluser_mc 1.432e-05 7.081e-05 3.295e+05 0.202 0.839696
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

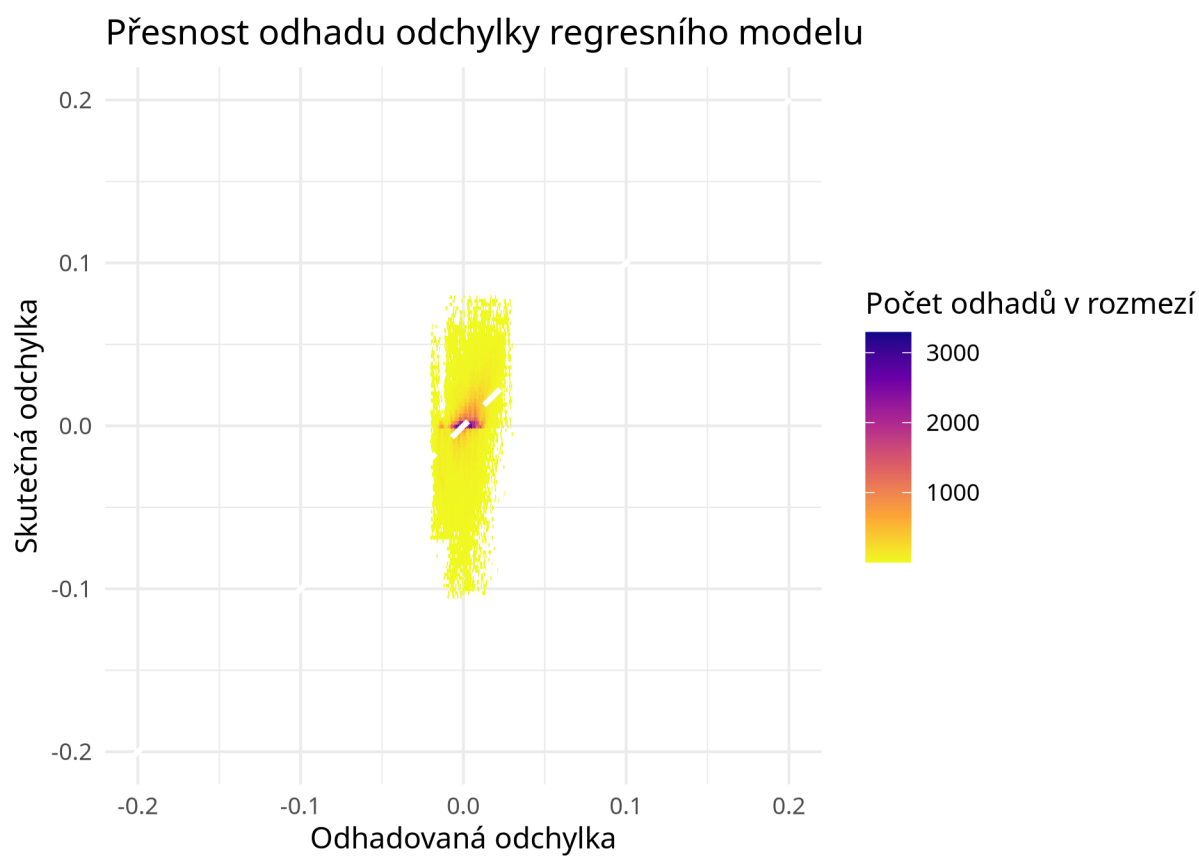
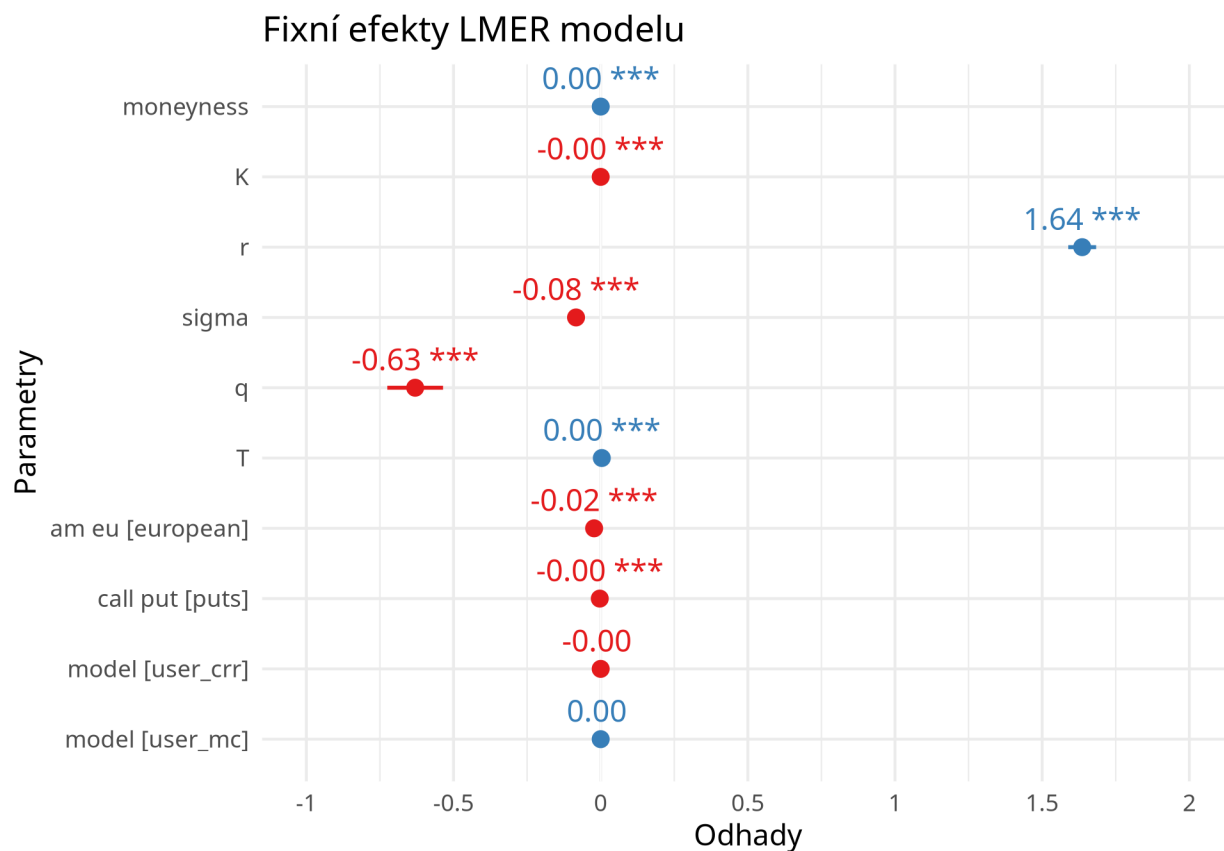
```

```

## Correlation of Fixed Effects:
##      (Intr) mnynss K      r      sigma q      T      am_rpn cll_pt
## moneyness -0.023
## K          -0.019  0.491
## r          -0.314  0.001 -0.023
## sigma      -0.197  0.012  0.023  0.314
## q          -0.139  0.013  0.009 -0.031  0.029
## T          -0.047 -0.100 -0.074  0.144  0.000 -0.001
## am_eueuropn -0.520 -0.001 -0.007  0.010  0.042  0.084  0.003
## call_putpts -0.007 -0.041  0.210 -0.015 -0.001  0.001 -0.017 -0.002
## modelsr_crr -0.011  0.000  0.000  0.000  0.001 -0.001  0.000  0.000  0.000
## modelusr_mc -0.011 -0.001 -0.001  0.000  0.002 -0.001  0.000  0.000  0.001
##      mdlr_c
## moneyness
## K
## r
## sigma
## q
## T
## am_eueuropn
## call_putpts
## modelsr_crr
## modelusr_mc  0.500
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
r2(rmr_relative)

## # R2 for Mixed Models
##
##   Conditional R2: 0.471
##   Marginal R2: 0.273

```



```
anova(rmr_relative)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##           Sum Sq Mean Sq NumDF  DenDF    F value    Pr(>F)
## moneyness 0.00488 0.00488      1 328991    17.7251 2.553e-05 ***
## K          0.48015 0.48015      1 318831 1743.7878 < 2.2e-16 ***
## r          1.26289 1.26289      1 328626 4586.5186 < 2.2e-16 ***
## sigma     2.07962 2.07962      1 307205 7552.6721 < 2.2e-16 ***
## q          0.04689 0.04689      1   7947  170.3037 < 2.2e-16 ***
## T          2.30629 2.30629      1 329492 8375.8952 < 2.2e-16 ***
## am_eu      0.00458 0.00458      1     15   16.6244 0.0009689 ***
## call_put   0.72165 0.72165      1 329488 2620.8642 < 2.2e-16 ***
## model      0.00003 0.00001      2 329486    0.0495 0.9517141
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranef(rmr_relative)
```

```
## $ticker
##           (Intercept)
## ^RUT      0.0200290083
## ^XDA     -0.0245611348
## ^XDB      0.0085831997
## ^XDE     -0.0057006223
## ^XDN      0.0016495493
## AAPL     -0.0124606424
## AMD       0.0124350572
## AMZN     -0.0044011312
## GOOGL     0.0026986683
## JNJ       0.0057295103
## KO       -0.0011771214
## META     -0.0026113541
## MSFT     -0.0036155649
## NFLX     -0.0033547402
## NVDA      0.0036175684
## PG        0.0002661214
## TSLA      0.0028736297
##
## with conditional variances for "ticker"
```

```
->
```